

Problem 2

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Problem 1 *State the form of the limit. Determine whether the form is determinate or indeterminate. Evaluate each limit.*

(a) $\lim_{x \rightarrow \infty} (\ln(1 + e^{-x}))^x$

(b) $\lim_{x \rightarrow \infty} \left(\frac{1}{x} + 1 \right)^{\frac{1}{x}}$

(c) $\lim_{x \rightarrow \infty} \left(\frac{\arctan(x)}{x} \right)$

(d) $\lim_{x \rightarrow \infty} (x - \ln(x))$

(e) $\lim_{x \rightarrow \infty} \left(x \ln \left(\frac{1}{x} \right) \right)$

(f) $\lim_{x \rightarrow 0^+} (\sin(x) \cot(x))$